

ADVANCED

Stage 6

Statistics (Adv), S3 Random Variables (Adv)

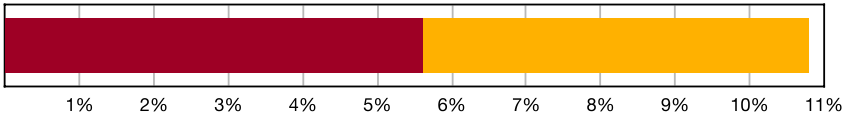
Normal Distribution (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 168 minutes (based on allocation of 1.5 minutes per mark)



S3 Random Variables



- Normal Distributions
- Probability Density Functions

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- S3 Random Variables has contributed a significant 10.8% per new syllabus Advanced exam since it was introduced in 2020.
- The topic is split into two sub-topics for the purposes of analysis: 1-Normal Distribution (5.6%) and 2-Probability Density Functions (5.2%).
- This analysis looks at Normal Distribution.

HSC ANALYSIS - What to expect and common pitfalls

- Normal Distribution (5.6%) has been examined in longer answer section of every new syllabus exam to date, as well as multiple choice questions in 2024 and 2020. It has the important characteristics of high mark allocations and low mean marks.
- The most common question type deals with z-score intervals that have featured in longer answer questions in every new syllabus exam to date.
- A deep understanding of the challenging z-score tables has been required in 3 of the last 4 exams and represents critical revision.
- The Advanced syllabus extends the Std2 syllabus to calculations involving the normal distribution probability density function. The syllabus explicitly mentions this in conjunction with the Trapezoidal Rule (review S3 EQ-Bank 2 and 17).
- Marker's Comments of note:** Graphically representing z-scores on a curve to calculate percentages has been specifically mentioned as a feature of the "more successful" responses in past HSC exams. Heed this advice by including a quick default sketch whenever possible and minimise errors!

Questions

1. Statistics, STD2 S5 2017 HSC 13 MC

The heights of Year 12 girls are normally distributed with a mean of 165 cm and a standard deviation of 5.5 cm.

What is the z-score for a height of 154 cm?

- A. -2
- B. -0.5
- C. 0.5
- D. 2

2. Statistics, 2ADV S3 2024 HSC 3 MC

Pia's marks in Year 10 assessments are shown. The scores for each subject were normally distributed.

	Pia's mark	Year 10 mean	Year 10 standard deviation
English	78	66	6
Mathematics	80	71	10
Science	77	70	15
History	85	72	9

In which subject did Pia perform best in comparison with the rest of Year 10?

- A. English
- B. Mathematics
- C. Science
- D. History

3. Statistics, 2ADV S3 SM-Bank 8 MC

The heights of females living in a small country town are normally distributed:

- 16% of the females are more than 160 cm tall.
- 2.5% of the females are less than 115 cm tall.

The mean and the standard deviation of this female population, in centimetres, are closest to

- A. mean = 135 standard deviation = 15
- B. mean = 135 standard deviation = 25
- C. mean = 145 standard deviation = 15
- D. mean = 145 standard deviation = 20

4. Statistics, STD2 S5 SM-Bank 4 MC

The length of a type of ant is approximately normally distributed with a mean of 4.8 mm and a standard deviation of 1.2 mm.

A standardised ant length of $z = -0.5$ corresponds to an actual ant length of

- A. 2.4 mm
- B. 3.6 mm
- C. 4.2 mm
- D. 5.4 mm

5. Statistics, STD2 S5 SM-Bank 1 MC

The head circumference (in cm) of a population of infant boys is normally distributed with a mean of 49.5 cm and a standard deviation of 1.5 cm.

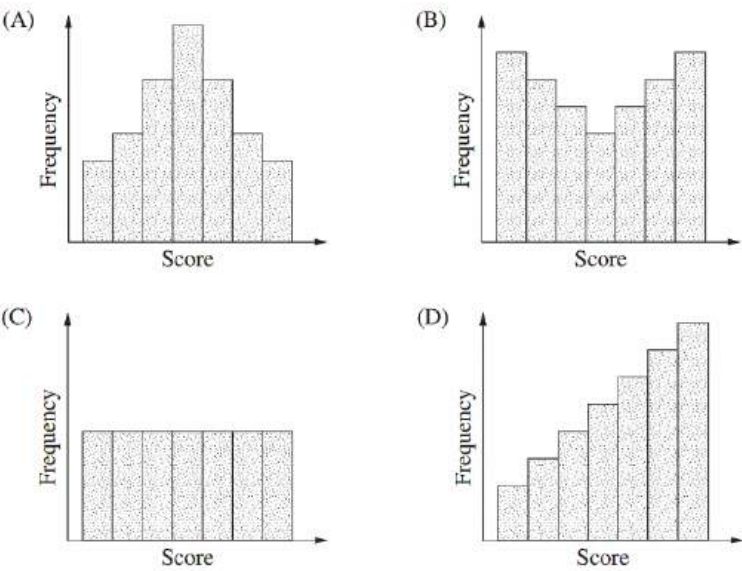
Four hundred of these boys are selected at random and each boy's head circumference is measured.

The number of these boys with a head circumference of less than 48.0 cm is closest to

- A. 3
- B. 10
- C. 64
- D. 272

6. Statistics, STD2 S5 2010 HSC 4 MC

Which of the following frequency histograms shows data that could be normally distributed?



7. Statistics, STD2 S5 2016 HSC 13 MC

The speed limit outside a school is 40 km/h. Year 11 students measured the speed of passing vehicles over a period of time. They found the set of data to be normally distributed with a mean speed of 36 km/h and a standard deviation of 2 km/h.

What percentage of the vehicles passed the school at a speed greater than 40 km/h?

- A. 2.5%
- B. 5%
- C. 47.5%
- D. 95%

8. Statistics, 2ADV S3 2020 HSC 3 MC

John recently did a class test in each of three subjects. The class scores on each test were normally distributed.

The table shows the subjects and John's scores as well as the mean and standard deviation of the class scores on each test.

Subject	John's score	Mean	Standard deviation
French	82	70	8
Commerce	80	65	5
Music	74	50	12

Relative to the rest of class, which row of the table below shows John's strongest subject and his weakest subject?

	Strongest subject	Weakest subject
A.	Commerce	French
B.	French	Music
C.	Music	French
D.	Commerce	Music

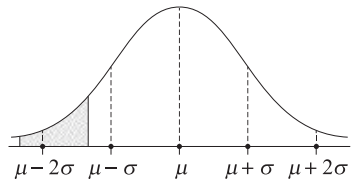
9. Statistics, 2ADV S3 2020 HSC 9 MC

Suppose the weight of melons is normally distributed with a mean of μ and a standard deviation of σ .

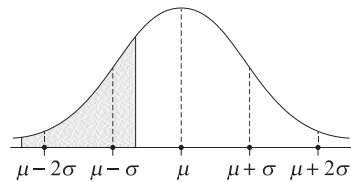
A melon has a weight below the lower quartile of the distribution but NOT in the bottom 10% of the distribution.

Which of the following most accurately represents the region in which the weight of this melon lies?

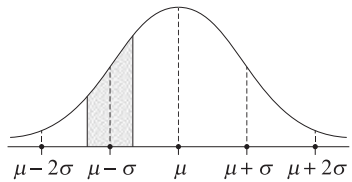
A.



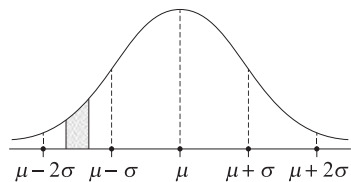
B.



C.



D.



10. Statistics, STD2 S5 2021 HSC 8 MC

On a test, Zac's mark corresponded to a z -score of 2. The test scores had a mean of 63 and a standard deviation of 8.

What was Zac's actual mark on the test?

- A. 65
- B. 67
- C. 73
- D. 79

11. Statistics, STD2 S5 SM-Bank 3 MC

The time, in hours, that each student spent sleeping on a school night was recorded for 1550 secondary-school students. The distribution of these times was found to be approximately normal with a mean of 7.4 hours and a standard deviation of 0.7 hours.

How many students would you expect to spend more than 8.1 hours sleeping on a school night?

You may assume for normally distributed data that:

- 68% of scores have z -scores between -1 and 1
- 95% of scores have z -scores between -2 and 2
- 99.7% of scores have z -scores between -3 and 3 .

- A. 16
- B. 248
- C. 1302
- D. 1510

12. Statistics, STD2 S5 SM-Bank 2 MC

The pulse rates of a large group of 18-year-old students are approximately normally distributed with a mean of 75 beats/minute and a standard deviation of 11 beats/minute.

The percentage of 18-year-old students with pulse rates less than 53 beats/minute or greater than 86 beats/minute is closest to

- A. 2.5%
- B. 5%
- C. 16%
- D. 18.5%

13. Statistics, STD2 S5 2006 HSC 17 MC

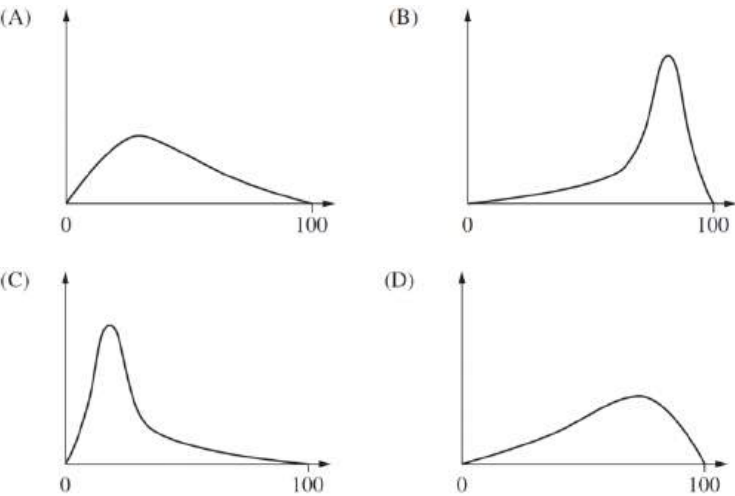
In a normally distributed set of scores, the mean is 23 and the standard deviation is 5.

Approximately what percentage of the scores will lie between 18 and 33?

- A. 34%
- B. 47.5%
- C. 68%
- D. 81.5%

14. Statistics, STD2 S1 2006 HSC 8 MC

Which of these graphs best represents positively skewed data with the smaller standard deviation?



15. Statistics, STD2 S5 2014 HSC 24 MC

The weights of 10 000 newborn babies in NSW are normally distributed. These weights have a mean of 3.1 kg and a standard deviation of 0.35 kg.

How many of these newborn babies have a weight between 2.75 kg and 4.15 kg?

- A. 4985
- B. 6570
- C. 8370
- D. 8385

16. Statistics, STD2 S5 2015 HSC 20 MC

A machine produces cylindrical pipes. The mean of the diameters of the pipes is 8 cm and the standard deviation is 0.04 cm.

Assuming a normal distribution, what percentage of cylindrical pipes produced will have a diameter less than 7.96 cm?

- A. 16%
- B. 32%
- C. 34%
- D. 68%

17. Statistics, STD2 S5 2018 HSC 23 MC

A set of data is normally distributed with a mean of 48 and a standard deviation of 3.

Approximately what percentage of the scores lies between 39 and 45?

- A. 15.85%
- B. 31.7%
- C. 47.5%
- D. 49.85%

18. Statistics, STD2 S5 2013 HSC 20 MC

There are 60 000 students sitting a state-wide examination. If the results form a normal distribution, how many students would be expected to score a result between 1 and 2 standard deviations above the mean?

You may assume for normally distributed data that:

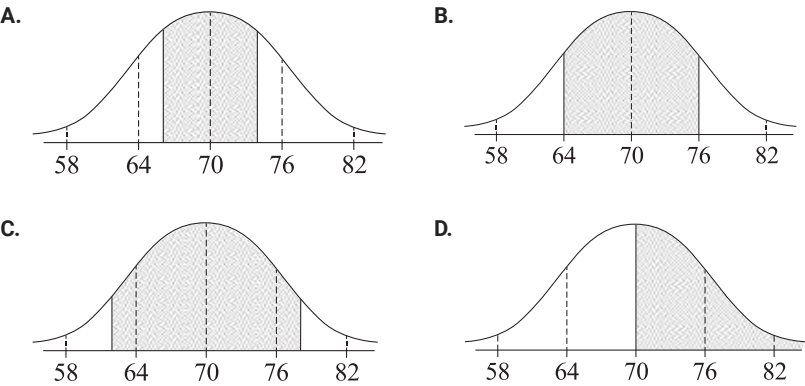
- 68% of scores have z -scores between -1 and 1
- 95% of scores have z -scores between -2 and 2
- 99.7% of scores have z -scores between -3 and 3 .

- A. 8100
- B. 16 200
- C. 20 400
- D. 28 500

19. Statistics, STD2 S5 2019 HSC 15 MC

The scores on an examination are normally distributed with a mean of 70 and a standard deviation of 6. Michael received a score on the examination between the lower quartile and the upper quartile of the scores.

Which shaded region most accurately represents where Michael's score lies?

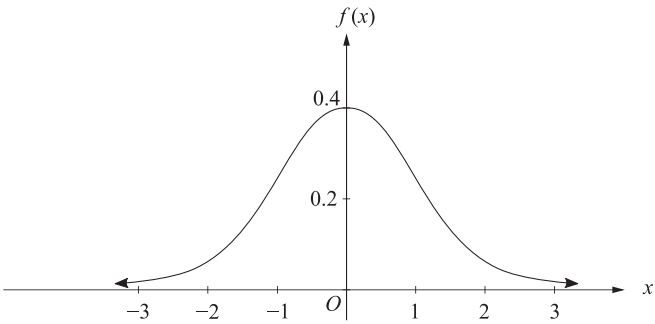


20. Statistics, STD2 S5 2018 HSC 27e

- Joanna sits a Physics test and a Biology test.
- i. Joanna’s mark in the Physics test is 70. The mean mark for this test is 58 and the standard deviation is 8.
Calculate the z -score for Joanna’s mark in this test. **(1 mark)**
- ii. In the Biology test, the mean mark is 64 and the standard deviation is 10.
Joanna’s z -score is the same in both the Physics test and the Biology test.
What is her mark in the Biology test? **(2 marks)**

21. Statistics, 2ADV S3 EQ-Bank 2

- Let X denote a normal random variable with mean 0 and standard deviation 1.
- The random variable X has the probability density function
- $$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \text{ where } x \in (-\infty, \infty)$$
- The diagram shows the graph of $y = f(x)$.



- i. Complete the table of values for the given function, correct to four decimal places. **(1 mark)**

$X = x$	-2	-1	0	1	2
$f(x)$		0.2420		0.2420	

- ii. Use the trapezoidal rule and 5 function values in the table in part i. to estimate

$$\int_{-2}^2 f(x) \, dx \quad \textbf{(2 marks)}$$

- iii. The weights of Rhodesian ridgebacks are normally distributed with a mean of 48 kilograms and a standard deviation of 6 kilograms.
Using the result from part ii., calculate the probability of a randomly selected Rhodesian ridgeback weighing less than 36 kilograms. **(2 marks)**

22. Statistics, 2ADV S3 2022 HSC 26

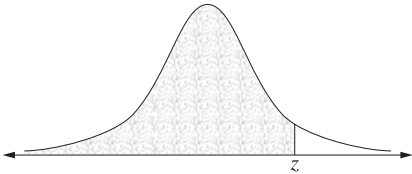
- The life span of batteries from a particular factory is normally distributed with a mean of 840 hours and a standard deviation of 80 hours.
- It is known from statistical tables that for this distribution approximately 60% of the batteries have a life span of less than 860 hours.
- What is the approximate percentage of batteries with a life span between 820 and 920 hours? **(3 marks)**

23. Statistics, 2ADV S3 2023 HSC 23

A random variable is normally distributed with a mean of 0 and a standard deviation of 1. The table gives the probability that this random variable lies below z for some positive values of z .

z	1.30	1.31	1.32	1.33	1.34	1.35	1.36	1.37	1.38	1.39
Probability	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

The probability values given in the table are represented by the shaded area in the following diagram.



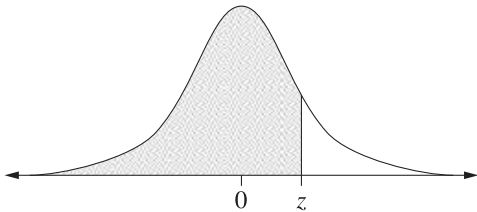
- The weights of adult male koalas form a normal distribution with mean $\mu = 10.40$ kg, and standard deviation $\sigma = 1.15$ kg.
- In a group of 400 adult male koalas, how many would be expected to weigh more than 11.93 kg? **(4 marks)**

24. Statistics, 2ADV S3 2024 HSC 23

A random variable is normally distributed with mean 0 and standard deviation 1. The table gives the probability that this random variable is less than z .

z	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4
<i>Probability</i>	0.7257	0.7580	0.7881	0.8159	0.8413	0.8643	0.8849	0.9032	0.9192

The probability values given in the table for different values of z are represented by the shaded area in the following diagram.



The scores in a university examination with a large number of candidates are normally distributed with mean 58 and standard deviation 15.

- a. By calculating a z -score, find the percentage of scores that are between 58 and 70. **(2 marks)**
- b. Explain why the percentage of scores between 46 and 70 is twice your answer to part (a). **(1 mark)**
- c. By using the values in the table above, find an approximate minimum score that a candidate would need to be placed in the top 10% of the candidates. **(2 marks)**

25. Statistics, 2ADV S3 EQ-Bank 17

The diastolic measurement for blood pressure in 35-year-old people is normally distributed, with a mean of 75 and a standard deviation of 12.

- i. A person is considered to have low blood pressure if their diastolic measurement is 63 or less. What percentage of 35-year-olds have low blood pressure? **(1 mark)**
- ii. Calculate the z -score for a diastolic measurement of 57. **(1 mark)**
- iii. The probability that a 35-year-old person has a diastolic measurement for blood pressure between 57 and 63 can be found by evaluating

$$\int_a^b f(x) \, dx$$

where a and b are constants and where

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- is the normal probability density function with mean 0 and standard deviation 1.
- By first finding the values a and b , calculate an approximate value for this probability by using the trapezoidal rule with 3 function values. **(3 marks)**
- iv. Hence, find the approximate probability that a 35-year-old person chosen at random has a diastolic measurement of 57 or less. **(1 mark)**

26. Statistics, STD2 S5 2007 HSC 25d

The results of two class tests are normally distributed. The means and standard deviations of the tests are displayed in the table.

	<i>Test 1</i>	<i>Test 2</i>
<i>Mean</i>	60	58
<i>Standard deviation</i>	6.2	6.0

- i. Stuart scored 63 in Test 1 and 62 in Test 2. He thinks that he has performed better in Test 1. Do you agree? Justify your answer using appropriate calculations. **(2 marks)**
- ii. If 150 students sat for Test 2, how many students would you expect to have scored less than 64? **(2 marks)**

27. Algebra, STD2 A2 2007 HSC 27b

- A clubhouse uses four long-life light globes for five hours every night of the year. The purchase price of each light globe is \$6.00 and they each cost \$ d per hour to run.
- i. Write an equation for the total cost (\$ c) of purchasing and running these four light globes for one year in terms of d . (2 marks)
 - ii. Find the value of d (correct to three decimal places) if the total cost of running these four light globes for one year is \$250. (1 mark)
 - iii. If the use of the light globes increases to ten hours per night every night of the year, does the total cost double? Justify your answer with appropriate calculations. (1 mark)
 - iv. The manufacturer's specifications state that the expected life of the light globes is normally distributed with a standard deviation of 170 hours.
What is the mean life, in hours, of these light globes if 97.5% will last up to 5000 hours? (1 mark)

28. Statistics, STD2 S5 2009 HSC 25d

- In Broken Hill, the maximum temperature for each day has been recorded. The mean of these maximum temperatures during spring is 25.8°C, and their standard deviation is 4.2° C.
- i. What temperature has a z -score of -1 ? (1 mark)
 - ii. What percentage of spring days in Broken Hill would have maximum temperatures between 21.6° C and 38.4°C?
You may assume that these maximum temperatures are normally distributed and that
 - 68% of maximum temperatures have z -scores between -1 and 1
 - 95% of maximum temperatures have z -scores between -2 and 2
 - 99.7% of maximum temperatures have z -scores between -3 and 3 . (3 marks)

29. Statistics, STD2 S5 2010 HSC 24c

- The marks in a class test are normally distributed. The mean is 100 and the standard deviation is 10.
- i. Jason's mark is 115. What is his z -score? (1 mark)
 - ii. Mary has a z -score of 0. What mark did she achieve in the test? (1 mark)
 - iii. What percentage of marks lie between 80 and 110?
You may assume the following:
 - 68% of marks have a z -score between -1 and 1
 - 95% of marks have a z -score between -2 and 2
 - 99.7% of marks have a z -score between -3 and 3 . (2 marks)

30. Statistics, STD2 S5 2011 HSC 27c

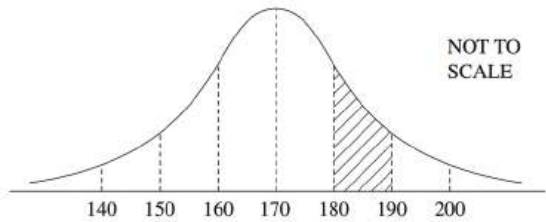
Two brands of light bulbs are being compared. For each brand, the life of the light bulbs is normally distributed.

Life of light bulbs (in hours)		
	Mean	Standard deviation
Brand A	450	25
Brand B	500	50

- i. One of the Brand B light bulbs has a life of 400 hours.
What is the z -score of the life of this light bulb? (1 mark)
- ii. A light bulb is considered defective if it lasts less than 400 hours. The following claim is made:
'Brand A light bulbs are more likely to be defective than Brand B light bulbs.'
Is this claim correct? Justify your answer, with reference to z -scores or standard deviations or the normal distribution. (2 marks)

31. Statistics, STD2 S5 2004 HSC 24c

The normal distribution shown has a mean of 170 and a standard deviation of 10.



- i. Roberto has a raw score in the shaded region. What could his z -score be? (1 mark)
- ii. What percentage of the data lies in the shaded region? (2 marks)

32. Statistics, STD2 S5 2005 HSC 26c

The weights of boxes of Brekky Bicks are normally distributed. The mean is 754 grams and the standard deviation is 2 grams.

- i. What is the z -score of a box of Brekky Bicks with a weight of 754 g? (1 mark)
- ii. What is the weight of a box that has a z -score of -1 ? (1 mark)
- iii. Brekky Bicks boxes are labelled as having a weight of 750 g. What percentage of boxes will have a weight less than 750 g? (2 marks)

33. Statistics, STD2 S5 2016 HSC 30d

The formula to calculate z -scores can be rearranged to give

$$\mu = x - \sigma z$$

where μ is the mean

x is the score

σ is the standard deviation

z is the z -score

- i. In an examination, Aaron achieved a score of 88, which corresponds to a z -score of 2.4.
Substitute these values into the rearranged formula above to form an equation. **(1 mark)**
- ii. In the same examination, Brock achieved a score of 52, which corresponds to a z -score of -1.2 .
Using this information, form another equation and solve it simultaneously with the equation from part (i) to find the values of μ and σ . **(3 marks)**

34. Statistics, 2ADV S3 2020 HSC 28

In a particular country, the hourly rate of pay for adults who work is normally distributed with a mean of \$25 and a standard deviation of \$5.

- a. Two adults who both work are chosen at random.
Find the probability that at least one of them earns between \$15 and \$30 per hour. **(3 marks)**
- b. The number of adults who work is equal to three times the number of adults who do not work.
One adult is chosen at random.
Find the probability that the chosen adult works and earn more than \$25 per hour. **(2 marks)**

35. Statistics, STD2 S5 2012 HSC 29b

A machine produces nails. When the machine is set correctly, the lengths of the nails are normally distributed with a mean of 6.000 cm and a standard deviation of 0.040 cm.

To confirm the setting of the machine, three nails are randomly selected. In one sample the lengths are 5.950, 5.983 and 6.140.

The setting of the machine needs to be checked when the lengths of two or more nails in a sample lie more than 1 standard deviation from the mean.

Does the setting on the machine need to be checked? Justify your answer with suitable calculations. **(2 marks)**

36. Statistics, STD2 S5 2013 HSC 29b

Ali's class sits two Geography tests. The results of her class on the first Geography test are shown.

58, 74, 65, 66, 73, 71, 72, 74, 62, 70

The mean was 68.5 for the first test.

- i. Calculate the standard deviation for the first test. Give your answer correct to one decimal place. **(1 mark)**
- ii. On the second Geography test, the mean for the class was 74.4 and the standard deviation was 12.4.
Ali scored 62 on the first test. Calculate the mark that she needed to obtain in the second test to ensure that her performance relative to the class was maintained. **(3 marks)**

37. Statistics, STD2 S5 2015 HSC 28b

The results of two tests are normally distributed. The mean and standard deviation for each test are displayed in the table.

	Mathematics	English
$\bar{x}$	70	75
s	6.5	8

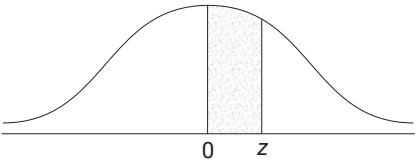
Kristoff scored 74 in Mathematics and 80 in English. He claims that he has performed better in English. Is Kristoff correct? Justify your answer using appropriate calculations. **(2 marks)**

38. Statistics, 2ADV S3 2021 HSC 22

A random variable is normally distributed with mean 0 and standard deviation 1. The table gives the probability that this random variable lies between 0 and z for different values of z .

z	0.1	0.2	0.3	0.4	0.5	0.6
Probability	0.0398	0.0793	0.1179	0.1554	0.1915	0.2257

The probability values given in the table for different values of z are represented by the shaded area in the following diagram.



- a. Using the table, find the probability that a value from a random variable that is normally distributed with a mean of 0 and standard deviation 1 lies between 0.1 and 0.5. **(1 mark)**
- b. Birth weights are normally distributed with a mean of 3300 grams and a standard deviation of 570 grams. By first calculating a z -score, find how many babies, out of 1000 born, are expected to have a birth weight greater than 3528 grams. **(3 marks)**

39. Statistics, 2ADV S3 2021 HSC 32

In a particular city, the heights of adult females and the heights of adult males are each normally distributed.

Information relating to two females from that city is given in Table 1.

Table 1

Height	Gender	Percentage of females in this city shorter than this person
175 cm	Female	97.5%
160.6 cm	Female	16%

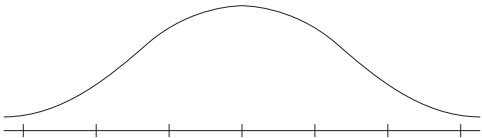
The means and standard deviations of adult females and males, in centimetres, are given in Table 2.

Table 2

	Mean	Standard deviation
Females	m	s
Males	$1.05m$	$1.1s$

A selected male is taller than 84% of the population of adult males in this city.

By first labelling the normal distribution curve below with the heights of the two females given in Table 1, calculate the height of the selected male, in centimetres, correct to two decimal places. **(4 marks)**



40. Statistics, STD2 S5 2019 HSC 38

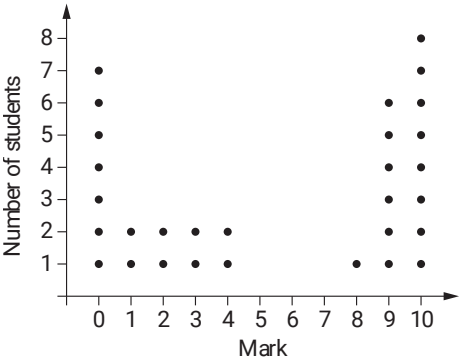
In a particular country, the birth weight of babies is normally distributed with a mean of 3000 grams. It is known that 95% of these babies have a birth weight between 1600 grams and 4400 grams.

One of these babies has a birth weight of 3497 grams. What is the z -score of this baby's birth weight? **(2 marks)**

41. Statistics, STD2 S5 2017 HSC 29d

All the students in a class of 30 did a test.

The marks, out of 10, are shown in the dot plot.



- i. Find the median test mark. **(1 mark)**
- ii. The mean test mark is 5.4. The standard deviation of the test marks is 4.22.
Using the dot plot, calculate the percentage of the marks which lie within one standard deviation of the mean. **(2 marks)**
- iii. A student states that for any data set, 68% of the scores should lie within one standard deviation of the mean. With reference to the dot plot, explain why the student's statement is NOT relevant in this context. **(1 mark)**

42. Statistics, STD2 S5 2008 HSC 28a

The following graph indicates z -scores of 'height-for-age' for girls aged 5 – 19 years.



- i. What is the z -score for a six year old girl of height 120 cm? **(1 mark)**
- ii. Rachel is $10\frac{1}{2}$ years of age.
 - (1) If 2.5% of girls of the same age are taller than Rachel, how tall is she? **(1 mark)**
 - (2) Rachel does not grow any taller. At age $15\frac{1}{2}$, what percentage of girls of the same age will be taller than Rachel? **(2 marks)**
- iii. What is the average height of an 18 year old girl? **(1 mark)**

For adults (18 years and older), the Body Mass Index is given by

$$B = \frac{m}{h^2} \text{ where } m = \text{mass in kilograms and } h = \text{height in metres.}$$

The medically accepted healthy range for B is $21 \leq B \leq 25$.

- iv. What is the minimum weight for an 18 year old girl of average height to be considered healthy? **(2 marks)**
- v. The average height, C , in centimetres, of a girl between the ages of 6 years and 11 years can be represented by a line with equation

$$C = 6A + 79 \text{ where } A \text{ is the age in years.}$$

- (1)** For this line, the gradient is 6. What does this indicate about the heights of girls aged 6 to 11? **(1 mark)**
- (2)** Give ONE reason why this equation is not suitable for predicting heights of girls older than 12. **(1 mark)**

Worked Solutions

1. Statistics, STD2 S5 2017 HSC 13 MC

$$\begin{aligned} z\text{-score} &= \frac{x-\mu}{\sigma} \\ &= \frac{154-165}{5.5} \\ &= -2 \end{aligned}$$

$\Rightarrow A$

2. Statistics, 2ADV S3 2024 HSC 3 MC

Consider the z-score of each option:

$$\begin{aligned} z\text{-score (English)} &= \frac{78-66}{6} = 2 \\ z\text{-score (Maths)} &= \frac{80-71}{10} = 0.9 \\ z\text{-score (Science)} &= \frac{77-70}{15} = 0.46\dots \\ z\text{-score (History)} &= \frac{85-72}{9} = 1.4\dots \end{aligned}$$

$\Rightarrow A$

3. Statistics, 2ADV S3 SM-Bank 8 MC

$$160 \rightarrow z\text{-score} = 1$$

$$115 \rightarrow z\text{-score} = -2$$

$$1 = \frac{160-\mu}{\sigma}$$

$$\mu + \sigma = 160 \dots (1)$$

$$-2 = \frac{115-\mu}{\sigma}$$

$$\mu - 2\sigma = 115 \dots (2)$$

$$(1) - (2)$$

$$3\sigma = 45$$

$$\sigma = 15$$

Substitute $\sigma = 15$ into (1)

$$\mu = 145$$

$\Rightarrow C$

Worked Solutions

4. Statistics, STD2 S5 SM-Bank 4 MC

$$\begin{aligned} z &= \frac{x-\mu}{\sigma} \\ -0.5 &= \frac{x-4.8}{1.2} \\ -0.6 &= x-4.8 \\ x &= 4.2 \text{ mm} \end{aligned}$$

$\Rightarrow C$

5. Statistics, STD2 S5 SM-Bank 1 MC

$$\mu = 49.5, \quad \sigma = 1.5$$

$$\begin{aligned} z\text{-score (49.5)} &= \frac{x-\mu}{\sigma} \\ &= \frac{48.0-49.5}{1.5} \\ &= -1 \end{aligned}$$

$\therefore$ Number of boys with a head under 48.0 cm

$$= 16\% \times 400$$

$$= 64$$

$\Rightarrow C$

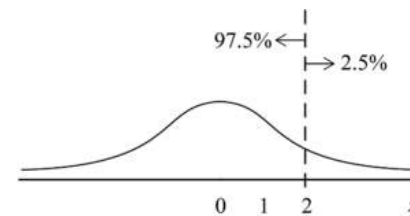
6. Statistics, STD2 S5 2010 HSC 4 MC

Normally distributed data have a frequency histogram graph that is shaped like a bell.

$\Rightarrow A$

7. Statistics, STD2 S5 2016 HSC 13 MC

$$\begin{aligned} z &= \frac{x-\mu}{\sigma} \\ &= \frac{40-36}{2} \\ &= 2 \end{aligned}$$



$\Rightarrow A$

8. Statistics, 2ADV S3 2020 HSC 3 MC

Calculate the z -score of each subject:

$$z\text{-score (French)} = \frac{82-70}{8} = 1.5$$

$$z\text{-score (Commerce)} = \frac{80-65}{5} = 3.0$$

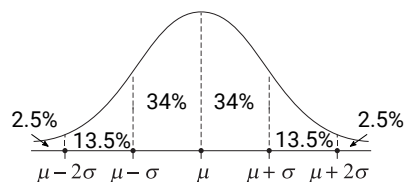
$$z\text{-score (Music)} = \frac{74-50}{12} = 2.0$$

$\therefore$ Commerce is strongest, French is weakest

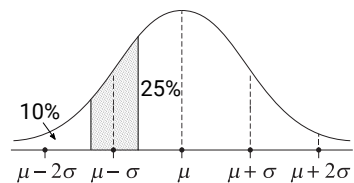
$\Rightarrow A$

9. Statistics, 2ADV S3 2020 HSC 9 MC

Distributions using μ and σ :



Adjusting the above to identify the interval $10\% < \text{weight} < 25\%$



$\Rightarrow C$

10. Statistics, STD2 S5 2021 HSC 8 MC

Method 1 (quicker)

$$\begin{aligned} \text{Actual mark} &= \mu + 2 \times \sigma \\ &= 63 + 2 \times 8 \\ &= 79 \end{aligned}$$

Method 2

$$\begin{aligned} z\text{-score} &= \frac{x-\mu}{\sigma} \\ 2 &= \frac{x-63}{8} \\ 16 &= x-63 \\ x &= 79 \end{aligned}$$

$\Rightarrow D$

11. Statistics, STD2 S5 SM-Bank 3 MC

Need to find z -score of 8.1 hours

$$\begin{aligned} z\text{-score} &= \frac{x-\mu}{\sigma} \\ &= \frac{8.1-7.4}{0.7} \\ &= 1 \end{aligned}$$

68% students with $-1 < z\text{-score} < 1$

$\therefore$ 16% have $z\text{-score} > 1$

$$\begin{aligned} \# \text{ Students} &= 16\% \times 1550 \\ &= 248 \end{aligned}$$

$\Rightarrow B$

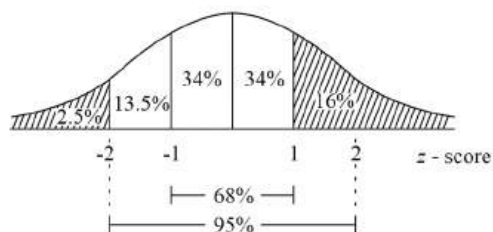
12. Statistics, STD2 S5 SM-Bank 2 MC

$$\mu = 75, \quad \sigma = 11$$

$$\begin{aligned} z\text{-score (53)} &= \frac{x - \mu}{\sigma} \\ &= \frac{53 - 75}{11} \end{aligned}$$

$$= -2$$

$$\begin{aligned} z\text{-score (86)} &= \frac{86 - 75}{11} \\ &= 1 \end{aligned}$$



From the diagram, the % of students that have a

z -score below -2 or above 1

$$= 2.5 + 16$$

$$= 18.5\%$$

$\Rightarrow D$

COMMENT: Two applications of the 68-95-99.7% rule are required. A good strategy is to *first draw a normal curve* and shade the required areas.

13. Statistics, STD2 S5 2006 HSC 17 MC

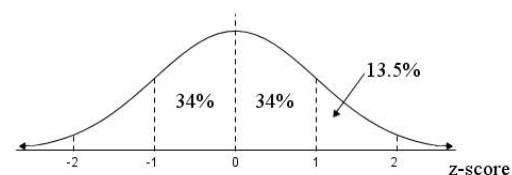
$$\mu = 23$$

$$\sigma = 5$$

$$\begin{aligned} z\text{-score (18)} &= \frac{x - \mu}{\sigma} \\ &= \frac{18 - 23}{5} \end{aligned}$$

$$= -1$$

$$\begin{aligned} z\text{-score (33)} &= \frac{33 - 23}{5} \\ &= 2 \end{aligned}$$



$\therefore$ Percentage between -1 and 2

$$= 34 + 34 + 13.5$$

$$= 81.5\%$$

$\Rightarrow D$

14. Statistics, STD2 S1 2006 HSC 8 MC

By elimination

Positive skew when the tail on the right side is longer.

$\therefore$ NOT B or D

A smaller standard deviation occurs when data is clustered more closely.

$\therefore$ NOT A where data is more widely spread.

$\Rightarrow C$

15. Statistics, STD2 S5 2014 HSC 24 MC

Find z -scores of 2.75 and 4.15 kg

◆ Mean mark 46%

$$z(2.75) = \frac{x - \mu}{\sigma} = \frac{2.75 - 3.1}{0.35} = -1$$

$$z(4.15) = \frac{4.15 - 3.1}{0.35} = 3$$

68% between $z = -1$ and 1

$\Rightarrow$ 34% between $z = -1$ and 0

99.7% between $z = -3$ and 3

$\Rightarrow$ 49.85% between $z = 0$ and 3

$\therefore$ % with z -scores between -1 and 3

$$= 34 + 49.85$$

$$= 83.85\%$$

$\therefore$ # Babies between 2.75 kg and 4.15 kg

$$= 83.85\% \times 10\,000$$

$$= 8385$$

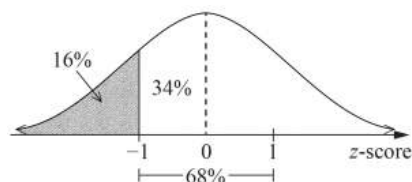
$\Rightarrow D$

16. Statistics, STD2 S5 2015 HSC 20 MC

$\mu = 8$ cm $s = 0.04$ cm

◆ Mean mark 50%.

$$\begin{aligned} z\text{-score}(7.96) &= \frac{x - \mu}{\sigma} \\ &= \frac{7.96 - 8}{0.04} \\ &= -1 \end{aligned}$$



$\therefore$ % of pipes with a diameter less than 7.96 cm.

$$= 50\% - 34\%$$

$$= 16\%$$

$\Rightarrow A$

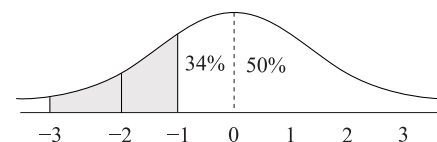
17. Statistics, STD2 S5 2018 HSC 23 MC

Given $\mu = 48$, $\sigma = 3$

◆ Mean mark 47%.

$$\begin{aligned} z\text{-score}(39) &= \frac{x - \mu}{\sigma} \\ &= \frac{39 - 48}{3} \\ &= -3 \end{aligned}$$

$$\begin{aligned} z\text{-score}(45) &= \frac{45 - 48}{3} \\ &= -1 \end{aligned}$$



$\therefore$ Scores between 39 and 45

$$\approx 16\%$$

$\Rightarrow A$

Note that % of scores below 39 is very small.

18. Statistics, STD2 S5 2013 HSC 20 MC

Since 68% between $z = 1$ and -1

$\Rightarrow$ 34% between $z = 0$ and 1

Likewise, 95% between $z = 2$ and -2

$\Rightarrow$ 47.5% between $z = 0$ and 2

$\therefore$ % between $z = 1$ and 2

$$= 47.5 - 34$$

$$= 13.5\%$$

$\therefore$ Students between $z = 1$ and $z = 2$

$$= 13.5\% \times 60\,000$$

$$= 8100$$

$\Rightarrow A$

19. Statistics, STD2 S5 2019 HSC 15 MC

68% of marks lie between 64 and 76 (mean $\pm 1\sigma$).

50% of marks lie between Q_1 and Q_3 .

◆◆◆ Mean mark 17%.

$\Rightarrow A$

20. Statistics, STD2 S5 2018 HSC 27e

i. $x = 70, \mu = 58, \sigma = 8$

$$\begin{aligned}\therefore \text{z-score} &= \frac{x-\mu}{\sigma} \\ &= \frac{70-58}{8} \\ &= 1.5\end{aligned}$$

ii. $1.5 = \frac{x-64}{10}$
 $x-64 = 15$
 $\therefore x = 79$

21. Statistics, 2ADV S3 EQ-Bank 2

i.

$X = x$	-2	-1	0	1	2
$f(x)$	0.0540	0.2420	0.3989	0.2420	0.0540

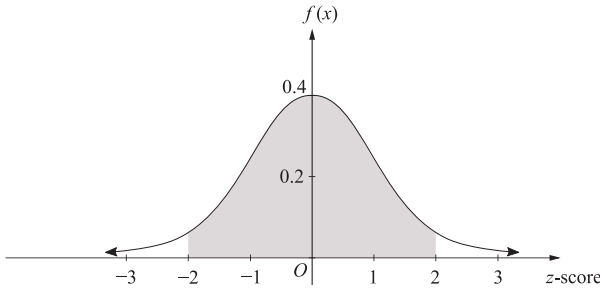
ii.

$X = x$	-2	-1	0	1	2
$f(x)$	0.0540	0.2420	0.3989	0.2420	0.0540
weight	1	2	2	2	1

$$\begin{aligned}\int_{-2}^2 f(x) &\approx \frac{1}{2} [0.0540 + 2(0.2420 + 0.3989 + 0.2420) + 0.0540] \\ &\approx \frac{1}{2} \times 1.8738 \\ &\approx 0.9369\end{aligned}$$

iii. $\mu = 48, \sigma = 6$

$$\text{z-score (36)} = \frac{x-\mu}{\sigma} = \frac{32-48}{6} = -2$$



Shaded area = 93.69%

By symmetry,

$$\begin{aligned}P(X < 36 \text{ kgs}) &= P(z < -2) \\ &= \frac{100-93.69}{2} \\ &= 3.155\%\end{aligned}$$

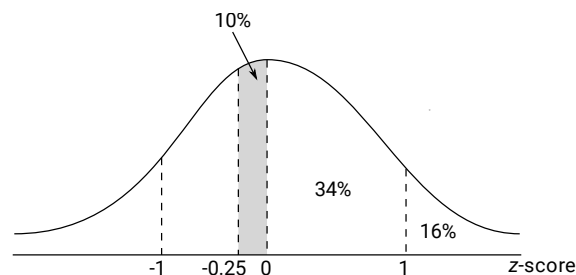
22. Statistics, 2ADV S3 2022 HSC 26

$$\mu = 840, \sigma = 80$$

$$z\text{-score}(860) = \frac{x - \mu}{\sigma} = \frac{860 - 840}{80} = 0.25$$

$$z\text{-score}(820) = \frac{820 - 840}{80} = -0.25$$

$$z\text{-score}(920) = \frac{920 - 840}{80} = 1$$



50% of batteries have a life span below 840 hours (by definition)

⇒ 10% lie between 840 and 860 hours

⇒ By symmetry, 10% lie between 820 and 840 hours

⇒ $P(-0.25 \leq z \leq 0) = 10\%$

∴ Percentage between 820 and 920

$$= P(-0.25 \leq z \leq 1)$$

$$= P(-0.25 \leq z \leq 0) + P(0 \leq z \leq 1)$$

$$= 10 + 34$$

$$= 44\%$$

23. Statistics, 2ADV S3 2023 HSC 23

$$\begin{aligned} z\text{-score}(11.93) &= \frac{x - \mu}{\sigma} \\ &= \frac{11.93 - 10.4}{1.15} \\ &= 1.330 \end{aligned}$$

$$P(\text{Koala weighs} > 11.93 \text{ kg}) = P(z > 1.330)$$

Using the table:

$$\begin{aligned} P(z > 1.33) &= 1 - 0.9082 \\ &= 0.0918 \end{aligned}$$

$$\begin{aligned} \therefore \text{Expected koalas} > 11.93 \text{ kg} &= 0.0918 \times 400 \\ &= 36.72 \\ &= 37 \text{ koalas*} \end{aligned}$$

*36 or 36.72 koalas would also receive full marks

24. Statistics, 2ADV S3 2024 HSC 23

$$\text{a. } z\text{-score}(58) = \frac{x - \mu}{\sigma} = \frac{58 - 58}{15} = 0$$

$$z\text{-score}(70) = \frac{70 - 58}{15} = 0.8$$

Using table:

$$\% \text{ between } 58-70 = 0.7881 - 0.5 = 0.2881 = 28.81$$

b. Normal distribution is symmetrical about the mean (58).

Since 70 and 46 are both the same distance (12) from the mean, the percentage of scores in this range will be twice the answer in part (a).

c. $z\text{-score } 1.3$ has a table value 0.9032

$$1 - 0.9032 = 0.0968 \Rightarrow \text{i.e. } 9.68\% \text{ of students score higher.}$$

Find x for a $z\text{-score}$ of 1.3:

$$\begin{aligned} 1.3 &= \frac{x - 58}{15} \\ x &= 1.3 \times 15 + 58 \\ &= 77.5 \end{aligned}$$

∴ Approx minimum score = 78%

25. Statistics, 2ADV S3 EQ-Bank 17

i. $z\text{-score (63)} = \frac{x-\mu}{\sigma}$
 $= \frac{63-75}{12}$
 $= -1$

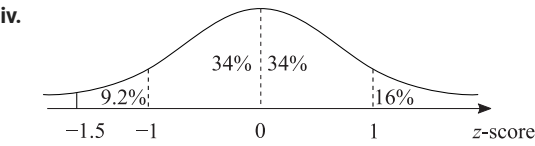
$\therefore$ 16% have low blood pressure

ii. $z\text{-score} = \frac{57-75}{12}$
 $= -1.5$

iii. $y = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

bp	57	60	63
x	-1.5	-1.25	-1
y	0.1295	0.1826	0.2420

Area = $\frac{h}{2}(y_0 + 2y_1 + y_2)$
 $\approx \frac{0.25}{2}(0.1295 + 2 \times 0.1826 + 0.2420)$
 ≈ 0.0920
 $\approx 9.2\%$



$P(\text{blood pressure} \leq 57) = 16-9.2$
 $\approx 6.8\%$

26. Statistics, STD2 S5 2007 HSC 25d

i. In Test 1, $\mu = 60, \sigma = 6.2$

$z\text{-score (63)} = \frac{x-\mu}{\sigma}$
 $= \frac{63-60}{6.2}$
 $= 0.483\dots$

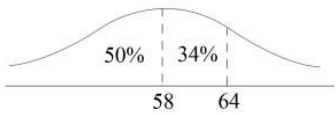
In Test 2, $\mu = 58, \sigma = 6.0$

$z\text{-score (62)} = \frac{62-58}{6.0}$
 $= 0.666\dots$

Since Stuart's z -score is higher in Test 2, his performance relative to the class is better despite his mark being slightly lower.

ii. In Test 2

$z\text{-score (64)} = \frac{64-58}{6}$
 $= 1$



$\Rightarrow 84\%$ have $z\text{-score} < 1$
 $\therefore$ # Students expected below 64
 $= 84\% \times 150$
 $= 126$

27. Algebra, STD2 A2 2007 HSC 27b

i. Purchase price = $4 \times 6 = \$24$

$$\begin{aligned}\text{Running cost} &= \# \text{ Hours} \times \text{Cost per hour} \\ &= 4 \times 5 \times 365 \times d \\ &= 7300d\end{aligned}$$

$$\therefore \$c = 24 + 7300d$$

ii. Given $\$c = \250

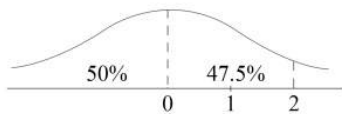
$$\begin{aligned}250 &= 24 + 7300d \\ 7300d &= 226 \\ d &= \frac{226}{7300} \\ &= 0.03095\dots \\ &= 0.031 \text{ \$/hr (3 d.p.)}\end{aligned}$$

iii. If d doubles to 0.062 \\$/hr

$$\begin{aligned}\$c &= 24 + 7300 \times 0.062 \\ &= \$476.60\end{aligned}$$

Since \$476.60 is less than $2 \times \$250$ (\$500), the total cost increases to less than double the original cost.

iv. $\sigma = 170$



z -score of 5000 hours = 2

$$\begin{aligned}z &= \frac{x - \mu}{\sigma} \\ 2 &= \frac{5000 - \mu}{170} \\ 340 &= 5000 - \mu \\ \mu &= 4660\end{aligned}$$

$\therefore$ The mean life of these globes is 4660 hours.

28. Statistics, STD2 S5 2009 HSC 25d

i. $\mu = 25.8$ $\sigma = 4.2$

$$\begin{aligned}\text{Using } z &= \frac{x - \mu}{\sigma} \\ -1 &= \frac{x - 25.8}{4.2} \\ x - 25.8 &= -4.2 \\ x &= 21.6^\circ\end{aligned}$$

$\therefore 21.6^\circ$ has a z -score of -1

ii. z -score of $21.6 = -1$

Find z -score of 38.4

$$z(38.4) = \frac{38.4 - 25.8}{4.2} = 3$$

68% of scores are between $z = -1$ and 1

$\Rightarrow$ 34% are between $z = -1$ and 0

99.7% of scores are between $z = -3$ and 3

$\Rightarrow$ 49.85% are between $z = 0$ and 3

$\therefore$ % Temps between 21.6° and 38.4°

$$\begin{aligned}&= 34\% + 49.85\% \\ &= 83.85\%\end{aligned}$$

♦ Mean mark 41%

MARKER'S COMMENT: Many students failed to use the answer to (d)(i), costing them valuable exam time.

29. Statistics, STD2 S5 2010 HSC 24c

- i. Given $\mu = 100$, $\sigma = 10$

If mark is 115,

$$\begin{aligned} z\text{-score} &= \frac{115 - \mu}{\sigma} \\ &= \frac{115 - 100}{10} \\ &= 1.5 \end{aligned}$$

- ii. $z\text{-score} = 0$ when mark equals the mean

$\therefore$ Mary's score was 100

iii. $z\text{-score of } 110 = \frac{110 - 100}{10} = 1$

$$z\text{-score of } 80 = \frac{80 - 100}{10} = -2$$

68% of marks lie between $z = -1$ and 1

$\Rightarrow$ 34% lie between $z = 0$ and 1

95% of marks lie between $z = -2$ and 2

$\Rightarrow$ 47.5% lie between $z = -2$ and 0

$\therefore$ % marks between 80 and 110

$$= 34\% + 47.5\%$$

$$= 81.5\%$$

MARKER'S COMMENT: Too many students had calculator errors in this question, giving away easy marks. BE CAREFUL!

♦ Mean mark 42%

30. Statistics, STD2 S5 2011 HSC 27c

- i. $z\text{-score of Brand B bulb (400 hrs)}$

$$\begin{aligned} &= \frac{x - \mu}{\sigma} \\ &= \frac{400 - 500}{50} \\ &= -2 \end{aligned}$$

- ii. $z\text{-score of Brand A bulb (400 hours)}$

$$\begin{aligned} &= \frac{400 - 450}{25} \\ &= -2 \end{aligned}$$

Since the $z\text{-score}$ for both brands is -2 , they are equally likely to be defective.

$\therefore$ The claim is incorrect.

♦ Mean mark 42%
MARKER'S COMMENT: A number of students found drawing normal curves in their solution advantageous.

31. Statistics, STD2 S5 2004 HSC 24c

i. $z\text{-score}(180) = \frac{x - \mu}{\sigma}$

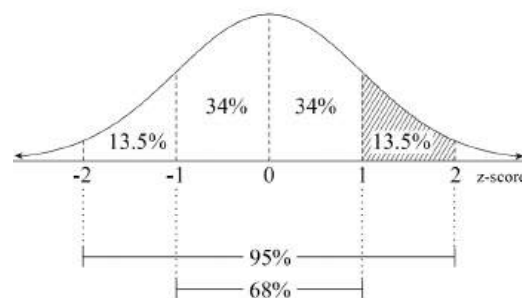
$$= \frac{180 - 170}{10}$$

$$= 1$$

$$\begin{aligned} z\text{-score}(190) &= \frac{190 - 170}{10} \\ &= 2 \end{aligned}$$

$$\therefore 1 \leq z\text{-score} \leq 2$$

- ii.



From the graph above,

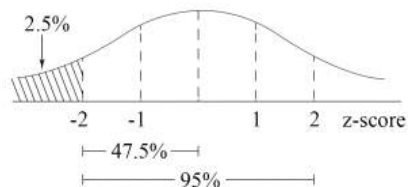
13.5% lies in the shaded area.

32. Statistics, STD2 S5 2005 HSC 26c

- i. z-score (754 g) = 0
(Noting that 754 g is the mean)

$$\begin{aligned}\text{ii. } z\text{-score} &= \frac{x - \mu}{\sigma} \\ -1 &= \frac{x - 754}{2} \\ x - 754 &= -2 \\ x &= 752 \text{ grams}\end{aligned}$$

$$\begin{aligned}\text{iii. } z\text{-score (750)} &= \frac{750 - 754}{2} \\ &= -2\end{aligned}$$



$\therefore$ Graph shows that 2.5% of boxes will weigh less than 750 g.

33. Statistics, STD2 S5 2016 HSC 30d

- i. $\mu = 88 - 2.4\sigma$
- ii. $\mu = 52 + 1.2\sigma \dots (1)$
 $\mu = 88 - 2.4\sigma \dots (2)$

Subtract (2)–(1):

$$0 = 36 - 3.6\sigma$$

$$3.6\sigma = 36$$

$$\therefore \sigma = 10$$

Substitute $\sigma = 10$ into (1):

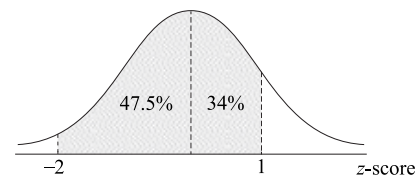
$$\begin{aligned}\therefore \mu &= 52 + 1.2 \times 10 \\ &= 64\end{aligned}$$

♦♦ Mean mark part (ii) 32%.
COMMENT: Recognise this is a simultaneous equation problem (now outside Std2 syllabus) and not a normal distribution one.

34. Statistics, 2ADV S3 2020 HSC 28

$$z\text{-score } (\$15) = \frac{x - \mu}{\sigma} = \frac{15 - 25}{5} = -2$$

$$z\text{-score } (\$30) = \frac{30 - 25}{5} = 1$$



Percentage of scores where $-2 \leq z \leq 1$
 $= 81.5\%$

$$\begin{aligned}P(\text{at least one earns between } \$15 - \$30) \\ &= 1 - P(\text{neither}) \\ &= 1 - (1 - 0.815)^2 \\ &= 1 - 0.185^2 \\ &= 0.965775\end{aligned}$$

$$\begin{aligned}\text{b. } P(\text{works}) &= \frac{3}{4}, \quad P(\text{earns} > \$25) = \frac{1}{2} \\ \therefore P(\text{works and earns} > \$25) \\ &= \frac{3}{4} \times \frac{1}{2} \\ &= \frac{3}{8}\end{aligned}$$

♦ Mean mark part (a) 40%.

35. Statistics, STD2 S5 2012 HSC 29b

$$\mu = 6.000, \quad \sigma = 0.040$$

Limits for $\pm 1 \sigma$:

$$6.000 + 0.040 = 6.040 \text{ (upper)}$$

$$6.000 - 0.040 = 5.960 \text{ (lower)}$$

Chosen nails: $5.950 \Rightarrow$ outside limits

$$5.983 \Rightarrow \text{inside}$$

$$6.140 \Rightarrow \text{outside}$$

Since 2 nails are outside 1σ limits

$\therefore$ Settings need to be checked.

♦ Mean mark 44%

36. Statistics, STD2 S5 2013 HSC 29b

i. $\sigma = 5.2201\dots$
 $= 5.2$ (to 1 d.p.)

♦ *Mean mark 39%*
COMMENT: Make sure you are confident with this function on your calculator!

ii. $z = \frac{x - \mu}{\sigma}$
 $z\text{-score (1st test)} = \frac{62 - 68.5}{5.2}$
 $= -1.25$

2nd test has z -score of -1.25 :

$$-1.25 = \frac{x - 74.4}{12.4}$$
$$x - 74.4 = -15.5$$
$$x = 58.9$$

$\therefore$ Ali needs to score 58.9

37. Statistics, STD2 S5 2015 HSC 28b

In Maths

$$z\text{-score}(74) = \frac{x - \mu}{\sigma}$$
$$= \frac{74 - 70}{6.5}$$
$$= 0.6153\dots$$

♦ *Mean mark 44%.*

In English

$$z\text{-score}(80) = \frac{80 - 75}{8}$$
$$= 0.625$$

$\Rightarrow$ Kristoff's z -score in English is higher than his z -score in Maths.
 $\therefore$ He is correct. He performed better in English.

38. Statistics, 2ADV S3 2021 HSC 22

a. $P(0.1 < x < 0.5) = 0.1915 - 0.0398$
 $= 0.1517$

♦♦ *Mean mark part (a) 29%.*
COMMENT: Note the Advanced and Std2 questions varied slightly but used the same table and graph.

b. $\mu = 330, \sigma = 570$

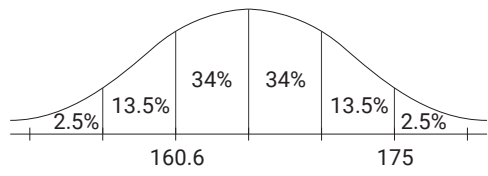
$$z\text{-score (3528)} = \frac{x - \mu}{\sigma}$$
$$= \frac{3528 - 3300}{570}$$
$$= 0.4$$

♦ *Mean mark part (b) 48%.*

$$P(z\text{-score} > 0.4) = 0.5 - 0.1554$$
$$= 0.3446$$

$$\therefore \text{Expected babies} > 3528 \text{ grams}$$
$$= 1000 \times 0.3446$$
$$= 344.6$$
$$\approx 345 \text{ babies}$$

39. Statistics, 2ADV S3 2021 HSC 32



$z\text{-score (175 cm, female)} = 2$
 $z\text{-score (160.6 cm, female)} = -1$

Find μ of female heights:

$$\mu - \sigma = 160.6$$
$$\mu + 2\sigma = 175$$
$$3\sigma = 175 - 160.6$$
$$\sigma = \frac{14.4}{3}$$
$$= 4.8 \text{ cm}$$
$$\therefore \mu = 165.4 \text{ cm}$$

Selected male's height has $z\text{-score} = 1$
$$\mu(\text{male}) = 1.05 \times 165.4 = 173.67$$
$$\sigma(\text{male}) = 1.1 \times 4.8 = 5.28$$

$$\therefore \text{Actual male height} = 173.67 + 5.28$$
$$= 178.95 \text{ cm}$$

◆ Mean mark 41%.

40. Statistics, STD2 S5 2019 HSC 38

95% babies within 1600–4400
i.e. $3000 \pm 2\sigma = 1600 - 4400$
$$2\sigma = 4400 - 3000$$
$$= 1400$$
$$\sigma = 700$$

$$\therefore z\text{-score (3497)} = \frac{x - \mu}{\sigma}$$
$$= \frac{3497 - 3000}{700}$$
$$= 0.71$$

◆ Mean mark 44%.

41. Statistics, STD2 S5 2017 HSC 29d

i. Median = $\frac{15\text{th} + 16\text{th score}}{2}$
$$= \frac{4 + 8}{2}$$
$$= 6$$

◆ Mean mark 50%.

ii. Lower limit = $5.4 - 4.22 = 1.18$
Upper limit = $5.4 + 4.22 = 9.62$
 $\therefore$ Percentage in between
$$= \frac{13}{30} \times 100$$
$$= 43.33 \dots$$
$$= 43\% \text{ (nearest \%)}$$

◆◆ Mean mark 34%.

iii. The statement assumes the data is normally distributed which is not the case here.

◆◆◆ Mean mark 13%.

i. $z\text{-score} = 1$

ii.(1) If $2\frac{1}{2}\%$ are taller than Rachel

$\Rightarrow z\text{-score of } +2$

$\therefore$ She is 155 cm

(2) At age $15\frac{1}{2}$, 155 cm has a $z\text{-score of } -1$

68% between $z = 1$ and -1

$\Rightarrow 34\%$ between $z = 0$ and -1

50% have $z \geq 0$

$\therefore$ % Above $z\text{-score of } -1$

$= 50 + 34$

$= 84\%$

$\therefore$ 84% of girls would be taller than Rachel at age $15\frac{1}{2}$.

iii. Average height of 18 year old has $z\text{-score} = 0$

$\therefore$ Average height = 163 cm

iv. $B = \frac{m}{h^2}$

$h = 163\text{ cm} = 1.63\text{ m}$

Given $21 \leq B \leq 25$, minimum healthy weight occurs when $B = 21$

$\Rightarrow 21 = \frac{m}{1.63^2}$

$m = 21 \times 1.63^2$

$= 55.794\dots$

$= 55.8\text{ kg (1 d.p.)}$

v.(1) It indicates that 6-11 year old girls, on average, grow

6 cm per year.

(2) Girls eventually stop growing, and the equation doesn't

factor this in.